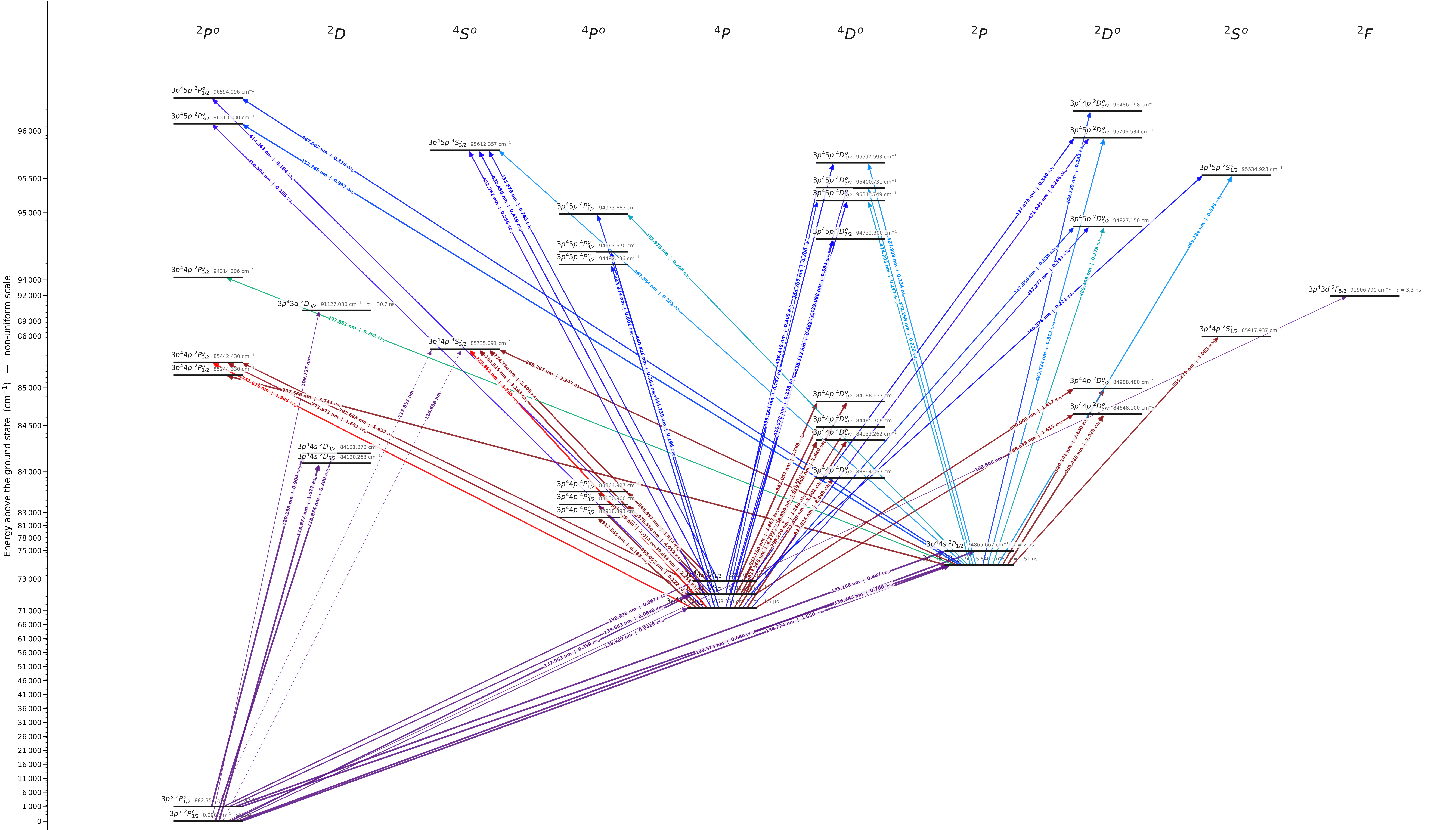
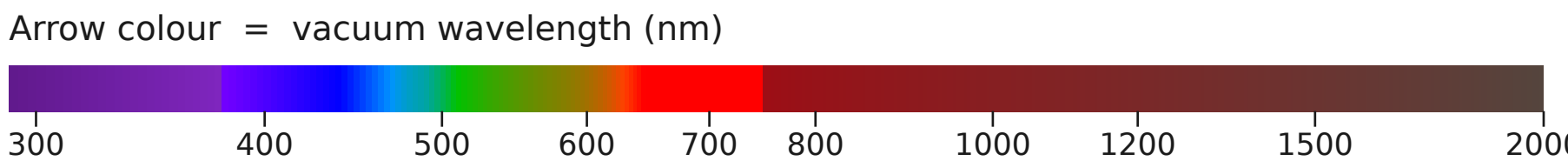


Chlorine

Cl (Cl I). Strongest classified transitions below 2 μm (NIST): 39 levels, 74 lines.

Cl⁺ ionisation limit 104591.01 cm⁻¹ = 12.96763 eV (95.611 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element [*J*||*er*||*J'*] in *e**a*₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient *A*. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3p ⁵ 2P _{3/2} ^o – 3p ⁴ 3d 2F _{5/2}	108.8059	–	2755.2962	–	–	48.23	–	interstellar Cl I 1088 Å line
3p ⁵ 2P _{3/2} ^o – 3p ⁴ 3d 2D _{5/2}	109.7369	–	2731.9196	–	–	5.184	–	interstellar Cl I 1097 Å line
3p ⁵ 2P _{3/2} ^o – 3p ⁴ 4p 4S _{3/2}	116.6384	–	2570.2734	–	–	–	–	two-photon LIF
3p ⁵ 2P _{3/2} ^o – 3p ⁴ 4p 4S _{3/2}	117.8512	–	2543.8211	–	–	–	–	TALIF detection of spin-orbit excited Cl
3p ⁵ 2P _{3/2} ^o – 3p ⁴ 4s 2P _{3/2}	134.7240	–	2225.2349	1.650	5.640e+08	105.4	85.2	134.72 nm VUV resonance line
3p ⁵ 2P _{1/2} ^o – 3p ⁴ 4s 2P _{3/2}	136.3447	–	2198.7827	0.700	9.780e+07	105.4	14.8	136.34 nm VUV line from the metastable 2P _{1/2} level

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

³⁵Cl hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3p ⁵ 2P _{3/2}	205.04687(3)	54.87290(6)	F=0: -700.34 F=1: -550.16 F=2: -194.94 F=3: +475.07
3p ⁵ 2P _{1/2}	1037.209(14)	–	F=1: -1296.51 F=2: +777.91

Measured A, B constants; sources in the data file.